MIL-S-80029D
20 June 1983
SUPERSEDING
MIL-S-80029C
25 October 1978

MILITARY SPECIFICATION

SANDER, DISK AND BELT

"This specification is approved for use by all Departments and Agencies of the Department of Defense."

1.1 Scope. This specification covers electric motor driven, cabinet and pedestal mounted, combination disk and belt sanders.

1.2 Classification. This specification covers sanders of the following sizes and classes. The size supplied shall be as specified (see 6.2.1).

Size 1 - 12 inch disk diameter - 6 inch belt width
Size 2 - 15 inch disk diameter - 6 inch belt width

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, addition, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this docu- or by letter.

FSC 3220

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-1188 - Industrial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawing, and the publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Officer, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by the manufacturer in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 79 - (ANSI C113.1) Electrical Standard for Metal Working Machine Tools and Plastics Machinery

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Ave., Boston, MA 02210.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

01.1 Woodworking Machinery, Safety Requirements for

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA) STANDARDS

MG-1 Motors and Generators
IC-S Industrial Controls and Systems

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L St., NW, Washington, DC 20037.)

(Technical Society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.).

2.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete sander for first article inspection and approval (see 6.2.1 and 6.3).

3.2 Design. The sander shall be one of the manufacturer's current models capable of sanding operations upon square, angular, and round workpieces in accordance with the requirements herein. The sander design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Controls. All controls shall be located convenient to the operator at his normal work station. Table adjustments for vertical, horizontal, and tilting shall provide easy manual control and shall remain locked in all adjusted positions.

3.2.2 Safety and health requirements. Cover, guards, or other safety devices shall be provided for all parts of the sander that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. In addition, the machine shall comply with all requirements of ANSI 01.1 that are designated therein as the responsibility of the machine manufacturer. In the event of conflict between the requirements of OSHA and the ANSI Standards, the requirements of OSHA shall apply. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

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3.3 Construction. The machine shall be constructed of parts which are without defects and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be used only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Scales. Scales indicating the degree of table tilt shall be graduated in increments not greater than one degree over the entire table tilt range. Scales shall be permanently and legibly engraved or etched on a non-glare background. Scale size shall be such that the graduations may be easily read from the normal operating position.

3.3.8 Plates. All words on instructions and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.3.9 Gears. All gears shall be machined from a material suitable for the intended purpose. All gears shall be heat treated by a process that will impart the hardness and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature wear.

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3.3.10 Reclaimed materials. The machine shall contain reclaimed materials to the maximum extent possible without jeopardizing its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use on the machine. Reclaimed materials shall include iron, steel, copper, brass, aluminum, fiber products, plastics and elastomers that have been collected from discarded solid, liquid, semi-solid, or gaseous waste such as garbage, refuse or sludge.

3.4 Components. The sander shall consist essentially of a frame and base, disk and belt assemblies, tables, drive, dust chute, electric motor and controls.

3.4.1 Frame and base. The frame and base of Size 1 sanders shall consist of an enclosed metal cabinet with an access door. The frame and base of Size 2 sanders shall be of the pedestal type construction. The frame and base shall have means for securely fastening the machine to the floor.

3.4.2 Disk sander assembly. The disk sander assembly shall consist basically of a disk and a tilting table.

3.4.2.1 Disk. The disk shall be constructed with the surface adjacent to the worktable finished for bonding an abrasive disk with an adhesive. The disk shall be balanced and mounted on a horizontal shaft supported by ball or roller bearings. The disk shall be securely mounted in a manner to facilitate installation, removal, and to maintain balance and alignment.

3.4.2.2 Disk table. The disk table workpiece support surface shall be flat within $\pm$0.005 inch per foot. The workpiece support surface shall normally be on a horizontal plane 90 degrees to the disk sanding surface and shall provide a tilt range conforming to the requirements of table I. The table shall be provided with a scale conforming to 3.3.7 which indicates table tilt, both up and down, off the horizontal plane.

3.4.3 Belt sander assembly. The belt sander assembly shall consist essentially of a driving drum, idler drum, an adjustable platen, and a table.

3.4.3.1 Belt drums. The driving drum and the idler drum shall be supported by ball or roller bearings which are sealed and lubricated for life. The belt drums shall be crowned to provide tracking alignment and shall be mounted within a frame having means of manual adjustment to maintain both belt tension and alignment. The frame shal be adjustable allowing sanding operations in both the horizontal and vertical positions.

3.4.3.2 Platen. A platen shall be mounted between the drums which will support the work area of the sanding belt. Means shall be provided for adjustment, and locking the platen in adjustment over its entire adjustment range.

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3.4.3.3 Belt sander table. The belt sander shall have a tilting table mounted to support a workpiece while the belt sander is in the vertical position. The table shall be constructed to tilt thru the table tilt range specified in table I. A scale conforming to 3.3.7 shall indicate the degrees of table tilt adjustment.

3.4.4 Drive. Unless otherwise specified (see 6.2.1), the sander shall be powered by either a direct or V-belt drive. On V-belt drive sanders, means shall be provided for adjusting belt alignment and tension.

3.4.5 Dust chutes. The drive drum of the sanding belt and the back and lower portion of the sanding disk shall be enclosed with a device that serves as a guard and dust chute. An outlet shall be shaped for connection to a dust collection system by means of an adapter.

3.4.6 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall be in accordance with this paragraph. The system shall be in accordance with the applicable requirements of National Electrical Manufacturers' Association (NEMA) Standards and National Fire Protection Association (NFPA) No. 79. Motors shall conform to NEMA MG-1 and their controls shall conform to NEMA IC-S. Each machine shall operate from a single 230/460, 3 phase, 60 Hz, AC power supply. The machines shall have a magnetic starter providing overload and undervoltage protection. A start-stop push button station shall be accessible from the operator's normal operating station. When specified (see 6.2.1), a foot operated start-stop switch shall also be installed on the sander.

3.5 Machine characteristics. The sizes and characteristics of the sanders shall meet the applicable requirements of table I.

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TABLE I. Characteristics and Sizes.

CHARACTERISTICS	REQUIREMENT	
	SIZE 1	SIZE 2
Disk:		
Diameter, inches	12	15
Speed, RPM	2400	1800
Table size, width x length, inches	9 x 16	9 x 21
Table tilt range, degrees:		
Up	20	20
Down	45	45
Belt:		
Width, inches	6	6
Platen size, width x length, inches	6 x 13	6 x 13
Speed, feet per minute	3400	3400
Table size, width x length, inches	7 x 14	7 x 14
Table tilt range, degrees:		
Up	20	20
Down	45	45
Motor:		
Horsepower	1	1

Requirements are not less than those shown.

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3.6 Performance. When tested in accordance with 4.6.2, the machine shall be capable of producing square and beveled edges in accordance with the accuracy requirements of table II. The sanded surfaces shall, in addition, be flat to an accuracy of 1/32 inch measured over any 6 inch span of the beveled edge. The accuracies are required when using the disk sander and the belt sander.

TABLE II. Performance Requirements.

Work table tilt settings	Resulting sanding angle (degrees)
0 Degrees (no tilt)	90 ± 1/2
10 Degrees up	80 ± 1/2
22 Degrees down	112 ± 1/2
45 Degrees down	135 ± 1/2

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), each sander shall be furnished with the following equipment:

a. One set of wrenches normally furnished with the machine by the manufacturer.

b. Either a one gallon can or one pound stick of adhesive (suitable for cementing the abrasive disks to the metal disk of the sander) for each machine.

3.8 Optional equipment. Optional equipment shall be as specified (see 6.2.1).

3.9 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate shall be securely attached to each sander. The nameplate shall contain the information listed below. If the sander is a special model, the model designation shall include the model of the basic standard sander and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase(s), frequency)
Disk diameter
Belt width
Date of manufacture
Contract Number or Order Number
National Stock Number or Plant Equipment Code

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3.10 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.11 Workmanship. Workmanship of the sander and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of the inspection requirements specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

.4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Quality conformance inspection shall consist of the examination in 4.5 and the test in 4.6.1. Failure of the item to pass the examination or test shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests.

4.6.1 Operational test. The sander shall be operated at no load for not less tahn 15 minutes. Proper operation of all controls, motors, adjusting mechanisms and other accessories shall be verified during the trial period.

4.6.2 Performance test. The sander shall be subjected to the following tests. The instruments and methods used in performing the tests shall be at the option of the contractor provided they accurately determine the tolerance of the sander. Sander performance shall conform to table II.

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4.6.1 Operational accuracy test. A one inch nominal thickness board having a width not less than 75 percent of the rated radius of the disk shall be used as the workpiece for all tests. The end grain of the workpiece shall be sanded to square and beveled edges, with the disk worktable adjusted, successively, to each of the angles specified in table II. The accuracy of the square edge, bevel angles, and the flatness of the sanded surfaces shall conform to all requirements of 3.6 and table II. The test shall be repeated using the belt sander in the vertical position, and with the abrasive belt table adjusted, successively, to the angles specified in table II. In all of the above tests, the workpiece may be rough sawn not closer than 5 degrees to the test angle prior to each sanding test.

4.7 Packaging inspection. Preservation, packing and marking shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing and marking shall conform to the requirements of ASTM-D-3951.

6. NOTES

6.1 Intended use. The sanders covered by this specification are intended for use in woodworking shops for sander finishing and polishing operation.

6.2 Ordering data.

6.2.1 Acquisition documents. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Size of sanding machine required (see 1.2).
c. First article approval, if required (see 3.1 and 6.3).
d. Specify additional safety and health requirements (see 3.2.2).
e. Painting, if different (see 3.3.5).
f. Drive, if different (see 3.4.4).
g. Electrical system, if different (see 3.4.6).
h. Foot operated start and stop switch, if required (see 3.4.6).
i. Standard equipment, if different (see 3.7).
j. Optional equipment, if required; fully describe (see 3.8).
k. Nameplate, if different (see 3.9).
l. First article inspection, if different (see 4.3).
m. Quality conformance inspection, if different (see 4.4).
n. Required level of preservation and packing (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operator manuals, parts lists, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on DD Form 1423 incorporated into the contract.

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6.3 First article. When a first article inspection is required, the end item will be tested and should be either a first article sample or a standard production sample from the contractors current inventory. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination and test approval of the documents first article.

6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment), the requesting activity is advised to analyze the existing hazard levels in the proposed operating environment and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of-operation guarding is required, the requesting activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(d).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians
Army - ME
Navy - SH
Air Force - 99

Preparing activity:
DLA - IP

Project Number:
3220-0137

Review activities:
Army - ME, CE
Navy - SH
Air Force - 84

User activities:
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-80029D	Military Specification (Sander, Disk and Belt)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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